DSA0602 DATA HANDLING AND VISUALIZATION | Class Activity 2: Relationship Workshop in R (35 Marks) | Slot C

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Scatterplots / Correlograms / PCA / Networks / Paired data / Bubble charts / Hexbin plots / 7 tasks x 5 marks / Tool: R (ggplot2 and the packages named in each question)

Instructions: Tool: R (ggplot2 and the packages named). Time: 3 hours. Total: 35 marks (7 tasks x 5). Every student does the same seven tasks; the data in the R code printed with each task are your own, so your numbers will differ from your neighbour's. Run each code block unchanged (do not edit the seed). Draw the chart asked for, then answer with specific values from YOUR output; generic statements earn no credit. Each chart needs a title, labelled axes with units and colour-blind-safe colours. Submit one R script, your charts and short answers as one file named RollNo_Name.

Relationship Workshop

Q1 [Scatterplots] (5 Marks) Scatterplot

Study weekly sales (Rs lakh) against footfall (visitors per day) across 24 weekly records.

```
set.seed(1215053)
n <- 24
sales <- round(rnorm(n, 85.0, 18.0), 1)
z <- (sales - 85.0) / 18.0
footfall <- round(900 + 200.0 * (-0.56 * z + 0.00 * z^2 + 0.35 * rnorm(n)), 0)
footfall[5] <- round(footfall[5] - 3.4 * 200.0, 0)   # one record was keyed in wrongly
df <- data.frame(sales, footfall)
```

Tasks:

1. Draw a scatterplot of sales (x) against footfall (y) with a straight-line fit (geom_smooth(method = "lm")) and a loess curve; label axes with units. [1.5]
2. Report the correlation r and the slope of the line, and describe the strength and direction in one sentence. [1]
3. Flag the outlier: give its row number and values, using the largest standardised residual (rstandard) from the straight-line fit. [1]
4. Fit a quadratic model (y ~ x + I(x^2)) and compare R-squared with the straight line; state whether a linear or non-linear trend fits better. [1]
5. Give one recommendation for the Sales Director that follows from the chart. [0.5]

Q2 [Correlograms] (5 Marks) Correlogram (correlation matrix heatmap)

Study the six measures recorded for 30 circles-periods at a telecom firm. The data frame df holds all six.

```
set.seed(1215163)
n <- 30
f1 <- rnorm(n)
f2 <- rnorm(n)
tower_density <- round(14.0 + 4.0 * (-0.91 * f1 + 0.41 * rnorm(n)), 1)
drop_rate <- round(2.6 + 0.7 * (0.90 * f1 + 0.43 * rnorm(n)), 1)
traffic_tb <- round(320 + 70.0 * (0.92 * f1 + 0.40 * rnorm(n)), 0)
users_lakh <- round(24.0 + 6.0 * (0.90 * f2 + 0.44 * rnorm(n)), 1)
churn_pct <- round(1.8 + 0.5 * (0.77 * f2 + 0.64 * rnorm(n)), 1)
arpu <- round(rnorm(n, 185, 25.0), 0)
df <- data.frame(tower_density, drop_rate, traffic_tb, users_lakh, churn_pct, arpu)
```

Tasks:

1. Compute cor(df), round to 2 decimals, and draw a correlogram (ggplot2 geom_tile with geom_text, or corrplot) with a diverging colour scale centred on 0. [1.5]
2. Identify the two most strongly correlated variable pairs (by absolute r) and report their r. [1.5]
3. Identify one surprising weak or negative correlation, report its r and explain in one sentence what it implies. [1]
4. Give one recommendation for the Network Head based on the matrix. [1]

Q3 [Dimension Reduction (PCA, t-SNE, UMAP)] (5 Marks) Dimension Reduction (PCA and t-SNE/UMAP)

Six measures were recorded for 36 destinations from three known groups (column grp).

```
set.seed(1215182)
n <- 36
grp <- rep(c("Group C", "Group A", "Group B"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)
footfall_k <- round(120.0 + 30.0 * (rnorm(n) + 0.90 * g), 1)
spend_rs <- round(3200 + 600.0 * (rnorm(n) + 0.40 * g), 0)
stay_nights <- round(2.8 + 0.7 * (rnorm(n) + 0.07 * g), 1)
rating10 <- round(7.6 + 0.8 * (rnorm(n) + 0.08 * g), 1)
occupancy_pct <- round(68.0 + 10.0 * (rnorm(n) + 1.30 * g), 1)
complaints <- round(4.0 + 1.2 * (rnorm(n) + 0.40 * g), 1)
df <- data.frame(footfall_k, spend_rs, stay_nights, rating10, occupancy_pct, complaints)
```

Tasks:

1. Run PCA on the scaled data (prcomp(df, scale. = TRUE)). Report the percentage of variance for PC1 and PC2 and the number of components needed for at least 80%. [1.5]
2. Plot PC1 against PC2 coloured by grp, then report the variable with the largest absolute loading on PC1. [1]
3. Judge how well PCA separates the groups using the share of variance in each of PC1 and PC2 explained by grp (R-squared of lm(score ~ grp)). [1]
4. Make a t-SNE (Rtsne) or UMAP plot of the same data (install the package if needed) and explain the variance-versus-neighbourhood trade-off between it and PCA; give one recommendation for the analyst. [1.5]

Q4 [Network Graphs] (5 Marks) Network Graph

Study the connections (flows of parcels delivered) between 8 hubs. Each row of the data is a link with a weight.

```
set.seed(1215212)
nodes <- c("Delhi", "Mumbai", "Pune", "Chennai", "Ahmedabad", "Hyderabad", "Bengaluru", "Kolkata")
pairs <- t(combn(nodes, 2))
u <- runif(nrow(pairs))
keep <- u < 0.33 | ((pairs[, 1] == "Bengaluru" | pairs[, 2] == "Bengaluru") & u < 0.95)
edges <- data.frame(from = pairs[keep, 1], to = pairs[keep, 2], weight = round(runif(sum(keep), 1, 20)))
```

Tasks:

1. Build the network with igraph (graph_from_data_frame). Draw it with node size proportional to degree and edge width proportional to weight, with node labels. [1.5]
2. Compute degree and betweenness for every node and name the most central node, justifying it with both numbers. [1.5]
3. Report the number of edges, the density (edge_density) and the number of connected components. [1]
4. Delete the most central node (delete_vertices) and report how many components remain; give one recommendation for the Operations Head. [1]

Q5 [Paired Data] (5 Marks) Paired / Slope Chart

The measure is staff on duty (higher is better). Values were recorded before and after a process change at 7 departments.

```
set.seed(1215352)
entity <- c("Paediatrics", "Radiology", "Cardiology", "Emergency", "Neurology", "Oncology", "Pharmacy")
n <- 7
before <- round(rnorm(n, 18, 2.0), 0)
change <- round(c(-1, 1, -1, 1, 1, -1, -1) * runif(n, 0.4, 1.4) * 2.4, 0)
change[5] <- 0.0   # one of them was almost unchanged
after <- before + change
df <- data.frame(entity, before, after)
```

Tasks:

1. Draw a paired slope chart (two points per entity joined by a line, geom_line(aes(group = entity))) with labelled endpoints. [1.5]
2. Classify each entity as improved, worsened or flat (flat means the absolute change is below 1% of the mean 'before' value = 0.180) and report the counts. [1]
3. Quantify the largest change (entity, size, and percentage of its 'before' value). [1]
4. Run a paired t-test (t.test(after, before, paired = TRUE)) and report t, p and the mean change; give one recommendation. [1.5]

Q6 [Bubble Charts] (5 Marks) Bubble Chart

Compare 7 genres on marketing spend (Rs lakh) (x), watch time (hours per user per month) (y) and subscribers (lakh) (bubble size).

```
set.seed(1215380)
entity <- c("Thriller", "Sci-Fi", "Animation", "Documentary", "Sports", "Comedy", "Drama")
n <- 7
z <- rnorm(n)
spend_lakh <- round(45.0 + 12.0 * z, 1)
watch_hrs <- round(11.0 + 3.0 * (0.90 * z + 0.40 * rnorm(n)), 1)
subs_lakh <- round(18.0 + 4.0 * rnorm(n), 1)
subs_lakh[5] <- round(18.0 * 2.4, 1); spend_lakh[5] <- round(45.0); watch_hrs[5] <- round(11.0)   # Sports has an unusual
subs_lakh
df <- data.frame(entity, spend_lakh, watch_hrs, subs_lakh)
```

Tasks:

1. Draw a bubble chart with geom_point(aes(size = ...)) and scale_size_area(), labels on every bubble, and a legend for size. [1.5]
2. Report the correlation between the x and y variables. [0.5]
3. Name the entity that is an outlier on the size variable even though its x/y position is near average, and show it with a z-score of size. [1.5]
4. State how many times larger its size value is than the median size, and explain why a plain scatterplot would have hidden it. [0.5]
5. Give one recommendation for the Content Head. [1]

Q7 [Hexbin Plots] (5 Marks) Hexbin Plot

A large paired dataset: loan amount (Rs lakh) (x) and applicant income (Rs thousand per month) (y), 3000 records.

```
set.seed(1215371)
n <- 3000
core <- runif(n) < 0.85
loan_lakh <- round(ifelse(core, rnorm(n, 38.0, 5.4), runif(n, 11.0, 65.0)), 1)
income_k <- round(ifelse(core, 52.0 - 0.70 * (loan_lakh - 38.0) / 5.4 + sqrt(1 - 0.49) * 7.2 * rnorm(n), runif(n, 16.0, 88.0)), 1)
df <- data.frame(loan_lakh, income_k)
```

Tasks:

1. Draw a hexbin plot (geom_hex(bins = 25) or the hexbin package) with a count legend and a viridis scale; also draw the plain scatterplot of the same data. [1.5]
2. Using a 10 x 10 grid (table(cut(x, 10), cut(y, 10))), find the busiest cell: give its x range, y range, count and share of all points. [1.5]
3. Report how many distinct (x, y) pairs there are (nrow(unique(df))) and explain why the scatterplot alone would mislead. [1]
4. Give one recommendation for the Zonal Manager. [1]

Code of Conduct: I certify that this work is my own. Signature: _____

Q1) Scatterplot

```
library(ggplot2)
set.seed(1215053)
n <- 24
Sales <- round(rnorm(n, 85, 18), 1)
z <- (sales - 85)/18
football <-
round(900 + 200 * (-0.56 * z + 0.35 * rnorm(n)), 0)
football[5] <- football[5] - 680
df <- data.frame(sales, football)

ggplot(df, aes(sales, football)) +
geom_point() + geom_smooth(method = "lm") +
geom_smoot(method = "loess", color = "red")

cor(sales, football)
m <- lm(football ~ sales, df)
rstandard(m)
summary(lm(football ~ sales + I(sales^2), df))
```

Q2) Correlogram

```
library(corrplot)
set.seed(1215163)
n <- 30
f1 <- rnorm(n); f2 <- rnorm(n)
tower_density <-
round(14 + 4 * (-0.91 * f1 + 0.41 * rnorm(n)), 1)
drop_rate <-
round(2.6 + 0.7 * (0.90 * f1 + 0.43 * rnorm(n)), 1)
traffic_tb <-
round(320 + 70 * (0.92 * f2 + 0.44 * rnorm(n)), 1)
users_lakh <-
```

```
round (24 + 6 * (0.90 * f2 + 0.44 * rnorm(n)), 1)
churn_pct <-
round (1.8 + 0.5 * (0.77 * f2 + 0.64 * rnorm(n)), 1)
arpu <- round (rnorm(n, 185, 25))
df <-
data. frame (tower_density, drop_rate, traffic_tb, users_
lakh, churn_pct, arpu)

C <- cor(df)
corrplot (c, method = "color", addcoef.col = "black")
Print (round (c, 2))
```

B3 PCA and t-SNE

```
library(ggplot2)
library (Rtsne)
set. seed (1215182)
n <- 36
grp <- rep(c("Group C", "Group A", "Group B"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)

football_k <- round(120 + 30 * (rnorm(n) + 0.90*g), 1)
spend_rs <- round (3200 + 600 * (rnorm(n) + 0.40*g))
stay_nights <- round (2.8 + 0.7 * (rnorm(n) + 0.07*g), 1)
rating_10 <- round (7.6 + 0.8 * (rnorm(n) + 0.08*g), 1)
occupancy_pct <- round (68 + 10 * (rnorm(n) + 1.30*g), 1)
complaints <- round (4 + 1.2 * (rnorm(n) + 0.40*g), 1)
df <-
data. fram (football_k, spend_rs, stay_nights, rati
ng10, occupancy_pct, complaints)

P <- prcomp (df, scale = TRUE)
summary (P)
plot( P$x[,1:2], col = as.factor(grp))
```

```
set.seed(1215182)
t<- Rtsne(scale(df), dims=2, perplexity=5)
plot(t$Y, col=as.factor(grp))
```

④ Network Graph

```
library(igraph)
set.seed(1215212)
nodes <-
c("Delhi", "Mumbai", "Pune", "Chennai", "Ahmedabad",
"Hyderbad", "Bengaluru", "Kolkata")
pairs <- t(combn(nodes,2))
u <- runif(nrow(pairs))
keep <- u<0.33] ((pairs[,1] == "Benglure")
pairs[,2] == "Bengaluru") & u<0.95)
edges <-
data.frame(from= pairs[keep, 1], to= pair[keep, 2]
,weight = round(runif(runif(sum(keep),1,20)))
g<-
graph_from_data_frame(edges, FALSE, nodes)
plot(g, vertex.size= degree(g)* 3+10, edge.width= E(g)$
weight/3)

degree(g)
betweenness(g)
ecount(g)
edge_density(g)
components(g)
g2 <-
delete_vertices(g, names(which.max(degree(g))))
plot(g2)
```

⑤ Paired Chart

```
library(ggplot2)
library(tidyr)
set.seed(1215352)
entity <-
c("Paediatrics", "Radiology", "Cardiology", "Emergency",
"Neurology", "Oncology", "Pharmacy")
 n <-7
before <- round(rnorm(n,18,2))
change <-
round(c(-1,1,-1,1,-1,-1,-1) * runif(n,0.4,1.4)* 2.4)
chang[5] <-0
after[5] before + change
 df <- data.frame( entity, before, after)

d <-
pivot_ longer (df, cols = c(before, after), name_to = "period",
values_to = "staff")
ggplot(d, aes( period, staff , group = entity , color = entity))+
  geom_ line()+ geom_point()
  +geom_ text (aes(label = staff), vjust = -1)

ifelse(abs(change) < 0.01 * mean( before), "Flat",
ifelse (change > 0 "Improved", "Worsened"))

t.test (after, before, paired = TRUE)
```

6. Bubble Chart

```
library(ggplot2)
set.seed(1215380)
entity <-
c("Thriller", "Sci-Fi", "Animation", "Documentary", "Spor
"Comedy", "Drama")
n <- 7
z <- rnorm(n)
spend_lakh <- round(45+12*z,1)
watch_hrs <-
round(11+3*(0.90*z+0.40*.rnorm(n)),1)
subs_lakh <- round(18+4*rnorm(n),1)
subs_lakh[5] <-round(18*2.4,1)
spend_lakh[5] <-45
watch_hrs[5] <-11
df <-
data.fram(entity, spend_lakh, watch_hrs, subs_lak
ggplot(df, aes(spend_lakh, watch_hrs, size = subs_lakh))+
geom_point()
+geom_text(aes(label= entity), vjust = -1)+ scale_size_area
cor(spend_lakh, watch_hrs)
scale(subs_lakh)
subs_lakh[which.max(scale(subs_lakh))]
subs_lakh[which.max(subs_lakh)]/
median(subs_lakh)
```

Hexbin Plot

```
library (ggplot2)
set.seed (1215371)
n <- 3000
core <- rnuif(n) <0.85
loan_lakh <-
round (ifelse (core, rnorm (n,38,5.4), runif(n,11,65)),1)
 income_k <-
round (ifelse (corne, 52-0.70* (loan_lakh- 38)/5.4+
sqrt (1-0.49) *7.2* rnorm(n), runif (n, 16,88)),1)
df <- data.frame (laon_lakh, income_k)
ggplot (df, aes(loan_lakh, income_k)) +
geom_hex(bins = 25) + scal- fill_viridis_c()

ggplot (df, aes(loan_lakh, income_k))+geom_point()

 x <- cut(df $loan_lakh,10)
 y <- cut (df $ income_k,10)
 tab <- table(x,y)
  print (tab)
 which ( tab == max( tab), arr.ind = TRUE)
 max(tab)
  max (tab) /n*100
 nrow(unique(df))
```